Machine Learning for Airbnb Price Prediction in Paris

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1 Introduction

Short-term rentals have become a major component of the Paris housing and tourism market, with tens of thousands of listings offered on Airbnb. For a host publishing a new listing, setting the right nightly price is a difficult task: the price must reflect the location, size and quality of the accommodation, while no booking history or guest reviews are yet available to guide the decision.

In this project, we use a dataset of Paris Airbnb listings published by Inside Airbnb, in which each listing is described by its location, its size, its amenities and the profile of its host. The main objective of this study is to predict the nightly price of a listing at the moment it goes online, using only the information available at that time.

To this end, we compare several machine learning methods of increasing complexity: simple baselines based on median prices, a regularised linear model (Ridge regression), a local method (k-nearest neighbours), and two tree ensembles, a random forest (bagging) and gradient boosting (XGBoost). Particular attention is paid to the evaluation protocol: since a single host often publishes many near-identical listings, the data are split by host, so that the models are always evaluated on listings from hosts they have never seen.

This approach allows us not only to compare the relevance and performance of these techniques, but also to better understand the factors that drive the price of a short-term rental in Paris. The remainder of this report is organised as follows. Section 2 describes the data, from the definition of the study population to an exploratory analysis. Section 3 details the methodology, Section 4 reports the results, Section 5 analyses the best model, and Section 6 concludes.

2 Data

2.1 Source

We use the Paris listings file published by Inside Airbnb [1], an independent project that collects publicly available information from the Airbnb website. The snapshot used in this study was collected on 16 September 2026 and is distributed under a CC BY 4.0 licence. The

*Code and data retrieval: `https://github.com/henribellamy/airbnb-paris-price-prediction`

raw file contains 77 149 listings described by 90 columns, covering the listing itself (location, accommodation, amenities, price), its host, its availability calendar and its guest reviews.

2.2 Selection of the variables

Our question, predicting the price of a listing at the moment it goes online, directly guides the choice of the variables: a column can only be used if its value is known when the listing is published. We therefore removed several groups of columns (Table 1).

First, we discarded columns that carry no usable information: 14 columns are entirely empty, and others contain URLs or scraping metadata. Columns containing personal data (host name, self-description, photos) were also removed. Second, we excluded all the information that only becomes available once the listing is online, such as guest reviews, review scores, availability and estimated occupancy. Using them would amount to *temporal leakage*: the model would rely on information that a new host cannot provide. Third, we removed the columns computed from the price itself (price quotes, estimated revenue), which would cause direct *target leakage*. Finally, redundant variables were dropped or merged; in particular, the number of beds was removed because it is strongly correlated with the guest capacity (Spearman $\rho = 0.82$) and accounts for most of the missing values.

Table 1: Selection of the variables.

Group removed	Reason	Columns left
–	raw file	90
Empty columns	no information	76
URLs, metadata, personal data	not informative, privacy	63
Reviews, availability, occupancy	unknown at listing time	42
Price quotes, estimated revenue	target leakage	36
Redundant variables, number of beds	duplicated information	22

From the remaining columns, we built features that the models can use directly. Boolean indicators were encoded as 0/1, and the textual description of the bathrooms was split into a number of bathrooms and a shared-bathroom indicator. The registration number, almost unique and therefore unusable as such, was replaced by three indicators describing the regulatory status of the listing (no registration, hotel exemption, inconsistent declaration). The list of amenities was summarised by its length and by ten binary indicators for amenities likely to affect the price (e.g. air conditioning, elevator, balcony). The 55 property types, many of them very rare, were grouped into six building types. Finally, the maximum stay was capped at 1 125 nights, the limit used by Airbnb, to remove technical placeholder values. The final dataset contains 30 explanatory variables.

2.3 Definition of the population

The study population was defined by successive filters (Table 2). The most important one concerns the price: 43.6 % of the listings have no price, most likely because they are inactive, and cannot be used since the price is our target. We then restricted the study to short-term rentals by keeping listings with a minimum stay of fewer than 30 nights; longer minimum stays correspond to medium-term leases, whose pricing logic differs. To remove implausible

prices, typically set by hosts to block their calendar rather than to rent, we kept listings priced between 14 € and 700 € per guest and per night, after a manual inspection of both tails of the distribution. Finally, we removed the few listings with a missing number of bedrooms or bathrooms, as well as four obvious input errors (e.g. 42 bathrooms for two guests).

Table 2: Definition of the population.

Step	Rule	Listings
Raw data	Inside Airbnb, 16/09/2026	77 149
Price available	target required	43 537
Short-term rentals	minimum stay < 30 nights	37 635
Plausible price	14 € $<$ price per guest < 700 €	37 416
Complete & consistent	bedrooms and bathrooms known, 4 outliers removed	**37 363**

The final population contains 37 363 short-term listings with a plausible price. Excluding listings without a price may introduce a selection bias, since inactive listings could differ from active ones; our conclusions therefore apply to listings that are currently offered for rent.

2.4 Exploratory analysis

Since each listing comes with its geographic coordinates (latitude and longitude), the listings can be plotted directly according to their position. Doing so reproduces the map of Paris (Figure 1): listings cover the whole city, with higher densities in the central and north-eastern arrondissements, and the most expensive ones (darker points) are concentrated in the centre and the west.

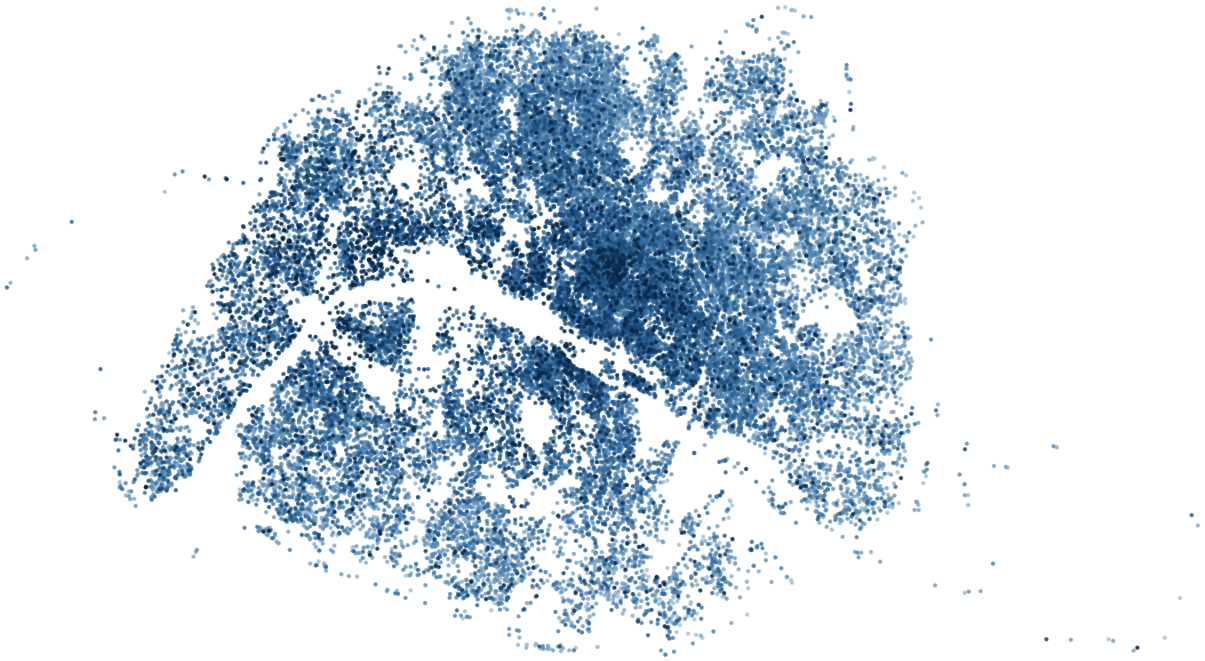


Figure 1: Location of the listings, coloured by nightly price (darker = more expensive).

The nightly price is strongly right-skewed: the median is about 239 €, half of the listings are priced between 160 and 381 €, while 3.5 % of them (1 298 luxury listings) exceed 1 000 € per night. On a logarithmic scale, the distribution becomes close to symmetric (Figure 2). We therefore model the logarithm of the price, which also makes the errors relative.

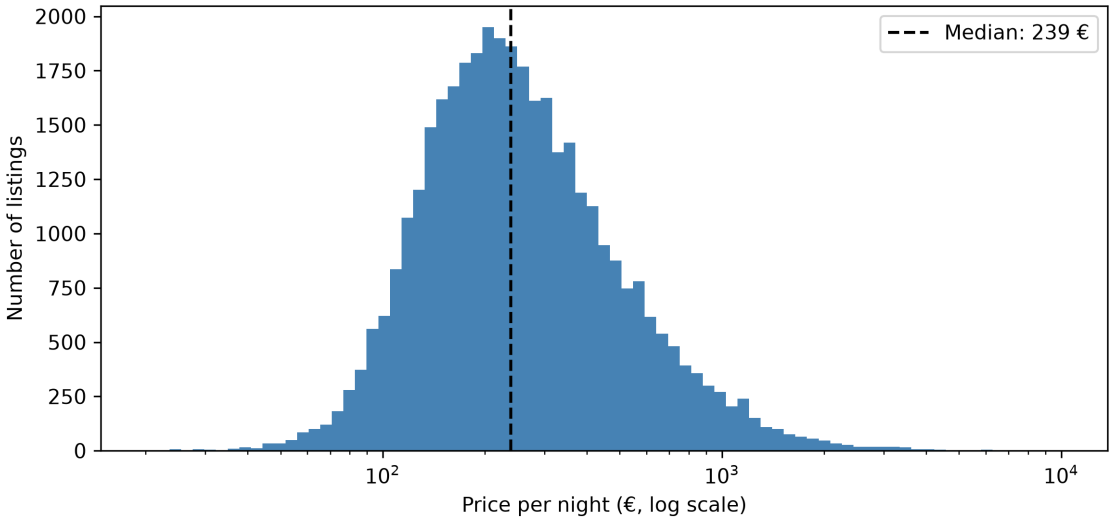


Figure 2: Distribution of the nightly price (logarithmic scale).

3 Methodology

The target is the logarithm of the nightly price. Models are selected on the root mean squared error on this scale (RMSE), which measures a relative error; we also report the mean absolute error in euros (MAE) and the coefficient of determination R^2.

Since a single host may publish dozens of near-identical listings (up to 227), a random split would let the models recognise listings rather than generalise. We therefore split the data *by host*: 20 % of the hosts form a test set, used only once for the final evaluation, and models are tuned and compared by 5-fold cross-validation [6] grouped by host on the remaining ones. All preprocessing (one-hot encoding, standardisation) is fitted inside a pipeline on the training folds only, using scikit-learn [7].

We compare models of increasing complexity: two baselines predicting the median price, globally or by arrondissement; Ridge regression [2] ($\alpha \approx 18$); k-nearest neighbours [3] ($k = 28$); a random forest [4], which averages deep trees trained on bootstrap samples (bagging); and XGBoost [5], which builds shallow trees sequentially, each correcting the previous ones (boosting; learning rate 0.03, depth 6, 600 trees). Hyperparameters were selected by grid search on the cross-validated RMSE.

4 Results

Model comparison. Table 3 reports the cross-validated performance of all models on the training set. The median by arrondissement barely improves on the global median, showing that location alone explains only a small share of the price. Ridge regression already reduces

the error strongly, k-nearest neighbours does slightly worse, and the tree ensembles perform best, as they capture interactions between variables. XGBoost achieves the lowest error; its advantage over the random forest is modest, of the order of one standard deviation, but it is also faster to train. We therefore select XGBoost as our final model.

Table 3: Cross-validation results on the training set (5 folds grouped by host; mean $\pm$ standard deviation over the folds for the RMSE).

Model	RMSE (log)	MAE (€)	R^2
Global median	0.682 ± 0.026	175	-0.01
Median by arrondissement	0.627 ± 0.021	163	0.14
Ridge regression	0.435 ± 0.008	118	0.59
k-nearest neighbours	0.466 ± 0.013	122	0.53
Random forest	0.408 ± 0.011	109	0.64
XGBoost	$\mathbf{0.397 \pm 0.011}$	**106**	**0.66**

Final evaluation. The selected model was refitted on the whole training set and evaluated once on the test set, together with the two baselines (Table 4). Its test error (0.39) is consistent with cross-validation, which indicates that the model selection did not overfit the validation folds. XGBoost reduces the error of the best baseline by about a third, with an average error of 102 € per night, and explains 63 % of the variance of the log price.

Table 4: Final evaluation on the test set (20 % of hosts, unseen during model selection).

Model	RMSE (log)	MAE (€)	R^2
Global median	0.649	163	-0.02
Median by arrondissement	0.606	153	0.11
XGBoost	**0.391**	**102**	**0.63**

Checking the grouping assumption. With a random split of the listings instead of a split by host, XGBoost would reach an RMSE of 0.374 ($R^2 = 0.70$). This gain is artificial, since near-identical listings of the same host would be seen during training: a naive protocol would overestimate the performance on new listings by about 6 %.

5 Analysis

The permutation importance of the variables [4] (Figure 3, right) measures how much the validation error increases when a variable is randomly shuffled. The price is mainly driven by the size of the accommodation (number of bedrooms, guest capacity, number of bathrooms) and by its location (latitude and longitude). Some amenities, such as air conditioning and a dishwasher, add a smaller premium, while the profile of the host plays a minor role. The random forest and XGBoost rely on largely the same variables, which suggests that these effects are robust to the choice of the model.

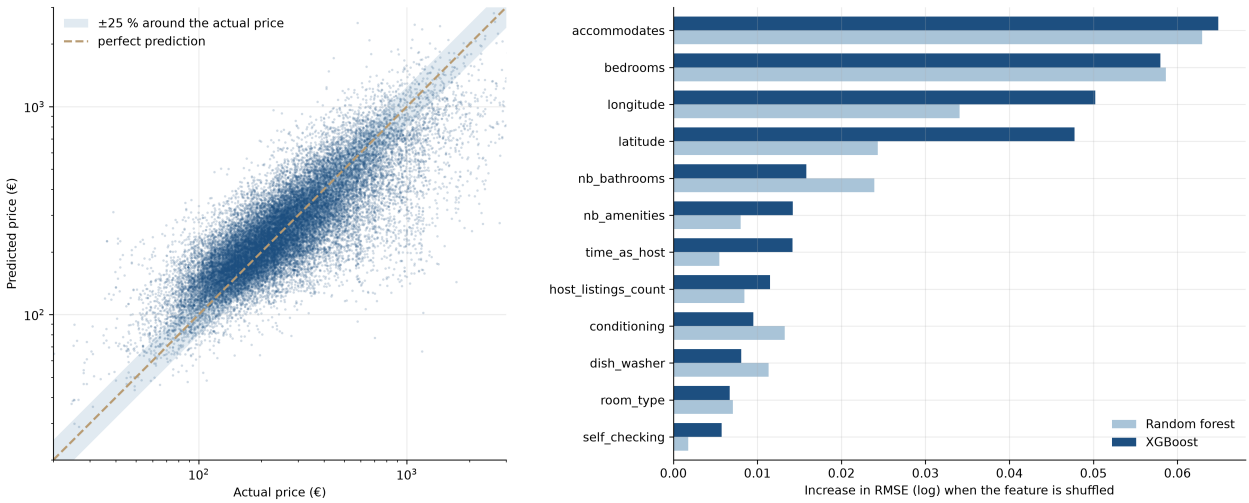


Figure 3: Left: predicted versus actual price for XGBoost (log scales). Right: permutation importance of the variables.

The comparison between predicted and actual prices (Figure 3, left) shows that the model captures the general trend but compresses the extremes: most listings fall within the $\pm 25\,\%$ band, but the cheapest ones are slightly overestimated and, above all, the most expensive ones are underestimated. This regression towards the mean is typical of a model that lacks the information needed to identify exceptional listings: the standing of luxury accommodations (renovation, view, decoration) is described in the text and photos rather than in the structured variables. Exploiting the description of the listings is therefore the most natural way to improve the model.

6 Conclusion

In this project, we tested a range of machine learning methods of increasing complexity, from simple baselines based on median prices to Ridge regression, k-nearest neighbours and two tree ensembles, a random forest and XGBoost. This diversity allowed us to assess the strengths and limits of each approach.

The best model, XGBoost, predicts the nightly price of a new listing with an RMSE of 0.39 on the log price for hosts it has never seen ($R^2 = 0.63$, average error of 102 €), reducing the error of the best baseline by about a third. Ridge regression already captures a large part of the price, but the tree ensembles perform better, as they account for interactions between variables. The price is mainly driven by the size and the location of the accommodation. This work also shows that the evaluation protocol matters as much as the model: since hosts publish many almost identical listings, splitting the data by host is necessary to estimate honestly the performance on new listings.

The main limit of our approach is the underestimation of luxury listings, whose standing is described in the text rather than in the structured variables. A natural improvement would be to use a language model to extract this information from the descriptions (view, renovation, standing), and add it to the existing features. With only the information available when a listing goes online, our model already provides a reliable price estimate, and this extension opens a promising path to make it even more accurate.

References

[1] Inside Airbnb. *Paris, 16 September 2026*. `https://insideairbnb.com/get-the-data/`. Licence CC BY 4.0.

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A Additional figures

Figure 4 complements the exploratory analysis of Section 2.4. Since each listing is assigned to an arrondissement, we computed the median nightly price of each one: it ranges from about 154 € in Ménilmontant, in the north-east, to about 397 € in Élysée, in the west, confirming the spatial pattern observed on the map of the listings.

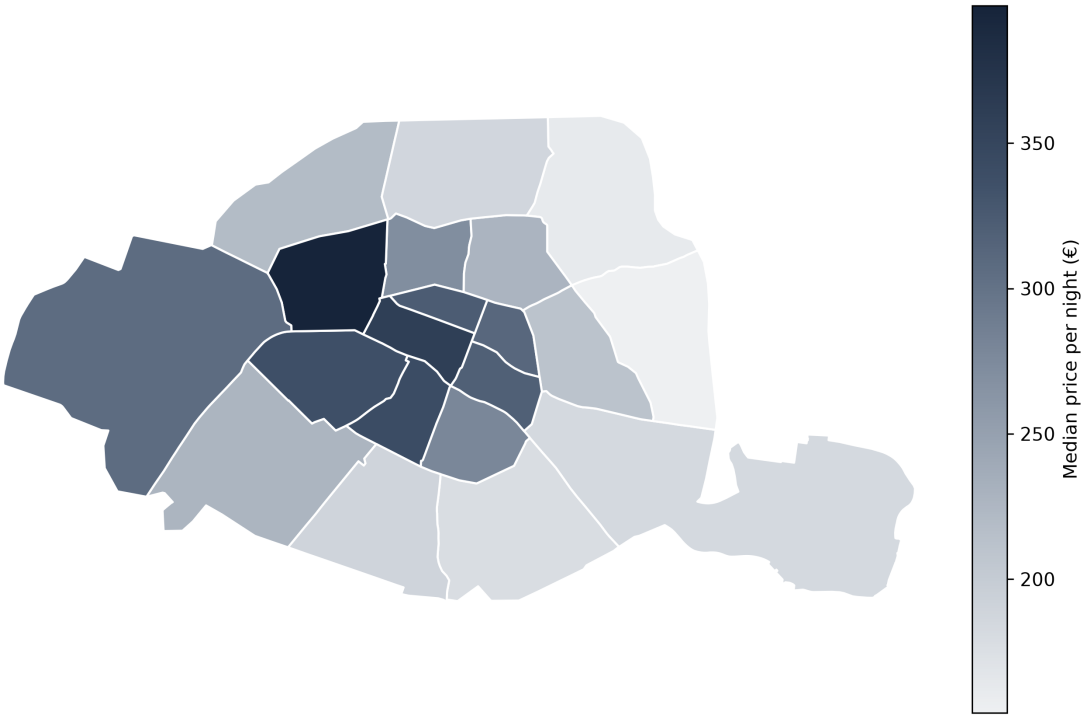


Figure 4: Median nightly price by arrondissement.